Bingjie Tang

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Summary

Senior Robotics Machine Learning Engineer at Tesla Optimus, working on learned dexterous manipulation for production humanoid robots—spanning data, policy training, evaluation, and on-robot deployment. PhD in Computer Science (USC), with a three-year research collaboration with NVIDIA Robotics on contact-rich, sim-to-real manipulation (IndustReal, AutoMate, FORGE, SRSA, Refinery).

Work Experience

- **Optimus AI, Tesla** *July 2025 - Present*
  Senior Robotics Machine Learning Engineer Palo Alto, USA
  - Own end-to-end delivery for the manipulation track: coordinate data-collection and labeling campaigns, train models hands-on, deploy policies to humanoid hardware, and operate evaluation at scale.
  - Develop core components of the model-development stack: data loading, training infrastructure, and debugging tooling.
  - Host the weekly cross-functional manipulation review, aligning hardware, firmware, sensing, simulation, data, and AI teams.
- **Seattle Robotics Lab, NVIDIA Corporation** *May 2022 - May 2025*
  Robotics Research Intern, advised by Yashraj Narang & Dieter Fox Seattle, USA
  - Three-year embedded research collaboration; led the contact-rich robotic assembly research line (IndustReal, AutoMate, FORGE, SRSA, MatchMaker, Refinery) from problem formulation through real-robot validation.
  - Released simulation environments, training code, and assets as open source in NVIDIA IsaacGymEnvs and Isaac Lab, including IndustRealKit and the 100-assembly AutoMate dataset.
  - Co-inventor on a granted US patent and two pending US patent applications covering RL training for robot control, skill retrieval, and automated asset generation.
- **Technology Engineering Group (TEG), Tencent** *Jun. 2017 - Sept. 2017*
  Software Engineer Intern Shenzhen, China

Patents and Publications

[1] **Bingjie Tang**, Iretiayo Akinola, Jie Xu, Bowen Wen, Michael Andres Lin, Dieter Fox, Gaurav S. Sukhatme, Fabio Ramos, Abhishek Gupta, Yashraj Narang. **Refinery: Active Fine-tuning and Deployment-time Optimization for Contact-Rich Policies.** *International Conference on Robotics & Automation (ICRA)*, 2026.

[2] Nicklas Hansen, Iretiayo Akinola, Yijie Guo, Jie Xu, **Bingjie Tang**, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang. **Generalizable Robotic Insertion with World Models.** *International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[3] Yijie Guo, **Bingjie Tang**, Iretiayo Akinola, Dieter Fox, Abhishek Gupta, Yashraj Narang. **SRSA: Skill Retrieval and Adaptation for Robotic Assembly Tasks.** *International Conference on Learning Representations (ICLR)*, 2025. **(Spotlight)** US Patent App. Pub. 2026/0091493.

[4] Yian Wang, **Bingjie Tang**, Chuang Gan, Dieter Fox, Kaichun Mo, Yashraj Narang, Iretiayo Akinola. **MatchMaker: Automated Robotic Assembly Asset Generation for Policy Learning in Simulation.** *International Conference on Robotics & Automation (ICRA)*, 2025. US Patent App. Pub. 2026/0080646.

[5] Michael Noseworthy, **Bingjie Tang**, Bowen Wen, Ankur Handa, Nicholas Roy, Dieter Fox, Fabio Ramos, Yashraj Narang, Iretiayo Akinola. **FORGE: Force-Guided Exploration for Robust Contact-Rich Manipulation under Uncertainty.** *IEEE Robotics and Automation Letters (RA-L)*, 2025.

[6] **Bingjie Tang**, Iretiayo Akinola, Jie Xu, Bowen Wen, Ankur Handa, Karl Van Wyk, Dieter Fox, Gaurav Sukhatme, Fabio Ramos, Yashraj Narang. **AutoMate: Specialist and Generalist Assembly Policies over Diverse Geometries.** *20th Robotics: Science and Systems (RSS)*, 2024.

[7] **Bingjie Tang***, Michael Lin*, Iretiayo Akinola, Ankur Handa, Gaurav Sukhatme, Fabio Ramos, Dieter Fox, Yashraj Narang. **IndustReal: Transferring Contact-Rich Assembly Tasks from Simulation to Reality.** *19th Robotics: Science and Systems (RSS)*, 2023. (*equal contribution) US Patent 12,552,021, granted 2026.

[8] **Bingjie Tang**, Gaurav S. Sukhatme. **Selective Object Rearrangement in Clutter.** *6th Annual Conference on Robot Learning (CoRL)*, 2022.

[9] **Bingjie Tang**, Matthew Corsaro, George Konidaris, Stefanos Nikolaidis, Stefanie Tellex. **Learning Collaborative Pushing and Grasping Policies in Dense Clutter.** *International Conference on Robotics & Automation (ICRA)*, 2021.

Open-Source Software & Datasets

- **Code:** IndustReal and AutoMate simulation environments, training code, and assets in NVIDIA IsaacGymEnvs / Isaac Lab.
- **Datasets:** IndustRealKit and the AutoMate dataset of 100 simulation- and real-world-compatible assemblies.

Education

- **University of Southern California** *Aug. 2020 - May 2025*
  PhD in Computer Science, advised by Gaurav S. Sukhatme Los Angeles, USA
- **Brown University** *Aug. 2018 - May 2020*
  MS in Computer Science, advised by George D. Konidaris & Stefanie A. Tellex. Providence, USA
- **Huazhong University of Science and Technology** *Aug. 2014 - May 2018*
  BS in Computer Science Wuhan, China

Professional Service

- **Associate Editor:** ICRA 2026.
- **Reviewer:** ICRA, IROS, CoRL, RSS, RA-L, T-RO.
- **Teaching:** TA for six course offerings at USC and Brown (deep learning, robotics, computer vision).

Skills

- **Programming Languages:** Python, C/C++.
- **Machine Learning:** PyTorch; large-scale GPU-parallel reinforcement learning.
- **Robotics & Simulation:** Isaac Lab, Isaac Gym; sim-to-real transfer and real-robot evaluation.